PROPOSITION LEFSCHETZ EXISTENCE IN ALGEBRAIZATION.TEX

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter algebraization.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate is from the extractor's stacks-candidates line, not from the earlier judged initial_corpus line; the H2 judgement and its reason below were made in this module by reading the author's own statement and complete proof in the fixed checkout. Original candidate key: stacks:algebraization.tex:proposition-lefschetz-existence.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 0889: sources/coherent.tex, source lines 6487–6487. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EJF: sources/algebraization.tex, source lines 6952–6952. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EKI: sources/more-morphisms.tex, source lines 14492–14492. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 0EKF: sources/more-morphisms.tex, source lines 14203–14203. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a

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Source reference. Original lemma 0B5T: sources/coherent.tex, source lines 4329–4329. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0887: sources/coherent.tex, source lines 6631–6631. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EIY: sources/algebraization.tex, source lines 5622–5622. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EJI: sources/algebraization.tex, source lines 7129–7129. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 0EKH: sources/more-morphisms.tex, source lines 14457–14457. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EHM: sources/properties.tex, source lines 2413–2413. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EL4: sources/algebraization.tex, source lines 9221–9221. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original proposition 0BXF: sources/coherent.tex, source lines 4226–4226. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EIQ: sources/algebraization.tex, source lines 4721–4721. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Reading revision r002. Original author context is supplied below and editorial reference bindings are qualified by the chapter. Mathematical statement and proof

are unchanged. Numbering is editorial; cited inputs remain in the complete fixed source.

Original author context (uncounted). Let k be a field. Let P be a proper scheme over k. Let $\mathcal{L}$ be an ample invertible $\mathcal{O}_P$-module. Let $s \in \Gamma(P, \mathcal{L})$ be a section and let $Q = Z(s)$ be the zero scheme, see Divisors, Definition 02OQ. Let $\mathcal{I} \subset \mathcal{O}_P$ be the ideal sheaf of Q. We will use $Coh(P, \mathcal{I})$ to denote the category of coherent formal modules introduced in Cohomology of Schemes, Section 0EHN.

Original author statement and proof.

Proposition 0.1. *In the situation above let $(\mathcal{F}_n)$ be an object of $Coh(P, \mathcal{I})$. Assume for all $q \in Q$ and for all primes $\mathfrak{p} \in \mathcal{O}^\wedge_{P,q}$, $\mathfrak{p} \notin V(\mathcal{I}^\wedge_q)$ we have*

$$depth((\mathcal{F}^\wedge_q)_\mathfrak{p}) + \dim(\mathcal{O}^\wedge_{P,q}/\mathfrak{p}) + \dim(\overline{\{q\}}) > 2$$

Then $(\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_P$-module.

Proof. By Cohomology of Schemes, Lemma 0889 to prove the lemma, we may replace $(\mathcal{F}_n)$ by an object differing from it by $\mathcal{I}$-torsion (see below for more precision). Let $T' = \{q \in Q \mid \dim(\overline{\{q\}}) = 0\}$ and $T = \{q \in Q \mid \dim(\overline{\{q\}}) \leq 1\}$. The assumption in the proposition is exactly that $Q \subset P$, $(\mathcal{F}_n)$, and $T' \subset T \subset Q$ satisfy the conditions of Lemma 0EJF with $d = 1$; besides trivial manipulations of inequalities, use that $V(\mathfrak{p}) \cap V(\mathcal{I}^\wedge_y) = \{\mathfrak{m}^\wedge_y\} \Leftrightarrow \dim(\mathcal{O}^\wedge_{P,q}/\mathfrak{p}) = 1$ as $\mathcal{I}^\wedge_y$ is generated by 1 element. Combining these two remarks, we may replace $(\mathcal{F}_n)$ by the object $(\mathcal{H}_n)$ of $Coh(P, \mathcal{I})$ found in Lemma 0EJF. Thus we may and do assume $(\mathcal{F}_n)$ is pro-isomorphic to an inverse system $(\mathcal{F}''_n)$ of coherent $\mathcal{O}_P$-modules such that $\text{depth}(\mathcal{F}''_{n,q}) + \dim(\overline{\{q\}}) \geq 2$ for all $q \in Q$.

We will use More on Morphisms, Lemma 0EKI and we will use the notation used and results found More on Morphisms, Section 0EKF without further mention; this proof will not make sense without at least understanding the statement of the lemma. Observe that in our case $A = \bigoplus_{m \geq 0} \Gamma(P, \mathcal{L}^{\otimes m})$ is a finite type k-algebra all of whose graded parts are finite dimensional k-vector spaces, see Cohomology of Schemes, Lemma 0B5T.

By Cohomology of Schemes, Lemma 0887 the pull back by $\pi|_U : U \to P$ is an object $(\pi|^*_U \mathcal{F}_n)$ of $Coh(U, f\mathcal{O}_U)$ which is pro-isomorphic to the inverse system $(\pi|^*_U \mathcal{F}''_n)$ of coherent $\mathcal{O}_U$-modules. We claim

$$\text{depth}(\pi|^*_U \mathcal{F}''_{n,y}) + \delta^Y_Z(y) \geq 3$$

for all $y \in U \cap Y$. Since all the points of Z are closed, we see that $\delta^Y_Z(y) \geq \dim(\overline{\{y\}})$ for all $y \in U \cap Y$, see Lemma 0EIY. Let $q \in Q$ be the image of y. Since the morphism $\pi : U \to P$ is smooth of relative dimension 1 we see that either y is a closed point of a fibre of π or a generic point. Thus we see that

$$\text{depth}(\pi^* \mathcal{F}''_{n,y}) + \delta^Y_Z(y) \geq \text{depth}(\pi^* \mathcal{F}''_{n,y}) + \dim(\overline{\{y\}}) = \text{depth}(\mathcal{F}''_{n,q}) + \dim(\overline{\{q\}}) + 1$$

because either the depth goes up by 1 or the dimension. This proves the claim.

By Lemma 0EJI we conclude that $(\pi|^*_U \mathcal{F}_n)$ canonically extends to X. Observe that

$$M_n = \Gamma(U, \pi|^*_U \mathcal{F}_n) = \bigoplus\nolimits_{m \in \mathbf{Z}} \Gamma(P, \mathcal{F}_n \otimes_{\mathcal{O}_P} \mathcal{L}^{\otimes m})$$

is canonically a graded A-module, see More on Morphisms, Equation (0EKH). By Properties, Lemma 0EHM we have $\pi|^*_U \mathcal{F}_n = \widetilde{M_n}|_U$. Thus we may apply Lemma 0EL4 to find a finite graded A-module N such that (M_n) and $(N/I^n N)$ are pro-isomorphic in the category of graded A-modules modulo A_+-torsion modules. Let $\mathcal{F}$

be the coherent $\mathcal{O}_P$-module associated to N, see Cohomology of Schemes, Proposition 0BXF. The same proposition tells us that $(\mathcal{F}/\mathcal{I}^n\mathcal{F})$ is pro-isomorphic to $(\mathcal{F}_n)$. Since both are objects of $Coh(P, \mathcal{I})$ we win by Lemma 0EIQ. $\square$